

Work Note v2 — Update

To: Prof. Taniike **Topic:** C₂-yield prediction for OCM — results under the stricter validation **Supersedes:** version 1. Problem setup, datasets and the five candidate methods are described there and are not repeated. This note reports only what has changed. **Code:** `ocm_eval.py`, `taniike_validation.py`, `phase3_lit_prior.py`, `phase4_family_diagnosis.py`, `phase5_target_audit.py`, `phase6_our_experiments.py`, `phase6_candidates.py`, `phase7_prep_ood.py`, `phase8_target_robustness.py`, `phase9_equal_effort_eval.py`, `phase10_condition_subsampling.py`

1. The improvement does not survive, and we withdraw it

Claim in v1	Status
PFT improves CV RMSE 10.6 % (1.907 vs 2.133)	Withdrawn — the gain was catalyst-identity leakage
Literature data helps the lab model	Not demonstrated in-domain ; it does help across preparation methods (§3)
Quantile normalisation is necessary	Not supported — label treatment moves RMSE inside run-to-run noise

Protocol	Baseline	PFT
Row-level split (v1)	2.118	1.912 (−9.7 %)
Catalyst-grouped split	2.943	2.995 (+1.8 %, worse)

Same models, same data; the difference between the two lines is leakage. **Mechanism:** our Stage-1 expert was trained on literature data *together with the lab training rows*, so under a random row split it had already seen the catalysts it was later asked to help predict. Catalyst-grouped splitting is now the default in our shared evaluation module and cannot be bypassed by accident.

2. What the data actually contains

The 89,074 measurements are **917 distinct catalysts** across 4,399 (catalyst, temperature) cells averaging 20.2 rows. Your description of these rows as measurements *under different reaction conditions* is borne out by the row counts, and we should have taken it more literally.

Cell sizes have a hard ceiling at exactly 27, a second at exactly **54 = 2 × 27**, nothing above; **15 catalysts hold exactly 135 rows, and all 15 decompose as (27, 27, 27, 27, 27)** — that is 5 temperatures × 27 condition settings = 135, matching your figure. We tested the competing reading that 27 was an export cut-off and rejected it: spacing between the two lowest values in a 27-row cell is 2.03× the interior spacing, against 0.90× when a larger cell is truncated to 27.

Because those 27 settings are absent from the feature table, **19.9 % of total yield variance lies within cells**, unreachable from composition and temperature, flooring row-level RMSE at **1.757** — a property of missing columns, not a physical limit. Version 1 reported 1.907, only 0.15 above that floor, which should itself have prompted suspicion. **Primary metrics are therefore catalyst-level, as you proposed:** rank correlation on maximum yield, and enrichment among top-ranked predictions.

3. Literature integration: null in-domain, useful across preparations

With the leak closed we retested four designs — a literature-only rank prior, similarity features, a gated prior, and a catalyst-level direct merge — with criteria fixed before running. **None improved on composition alone** (control $p = 0.761$; best variant 0.758). A follow-up across **all 28 element families with ≥50 catalysts** was also null: mean effect −0.0025, strongest correlation with any coverage measure $|p| = 0.276$ against a pre-registered threshold of 0.5.

The exception is cross-preparation transfer. Predicting *non-impregnation* literature from lab data alone gives $p = 0.238$ with enrichment 0.42× — worse than random selection. Adding impregnation literature to training raises this to **$p = 0.388$, enrichment 1.34×**. This is the one setting where literature data measurably helps, and it is where the lab has no coverage at all. Two caveats: absolute performance remains poor, and a **plain merge outperforms our two-stage construction (0.388 vs 0.318)** — the value is the data, not the machinery.

4. Screening performance, and a correction to our own number

Metric	All 917 catalysts	Equal-effort set (771)
Rank correlation (predicted vs observed max)	0.767	0.724
Enrichment of true top-decile	4.35×	3.77×
Precision@20	0.45	0.35

Why the second column exists. Grid coverage is coupled to performance — cells run further contain better yields — so a score over all 917 is partly a record of which experiments were completed. The equal-effort set is the 771 catalysts with ≥ 20 measurements in at least one cell; there the coupling is gone by measurement, with Spearman(measurement count, observed maximum) falling from **+0.293 to +0.003**. We regard 0.724 as the honest figure. The drop is not an artifact of scoring fewer catalysts: 300 **random** 771-catalyst subsets of the same predictions give 0.767 with a 95 % range of 0.756–0.780, and 0.724 lies below it.

A negative control we value more than either number. Refitting the identical model with the *number of measurements* as its target — it never sees a yield — still reaches $\rho = \mathbf{0.400}$ against observed maximum yield, but enrichment **0.87×**, no better than chance. Rank correlation is partly purchasable from experimental effort; enrichment is not. That is why enrichment is our primary metric.

One limit for campaign design. Inside the model's own top-ranked region the internal ordering carries little information: $\rho = 0.179$ within the top 150, **−0.066 within the top 20**. It *selects* well (top 20 average 17.3 % observed maximum against ~10.5 % library-wide) but does not *order* within its selection — a shortlist is a set to test, not a ranking.

5. Training label: maximum versus median

Each catalyst's label must be built from its ~135 readings. We compared the single highest reading anywhere against the median *within* each temperature group — across the cluster of near-identical readings — then the best of those per-group medians. Both scored against the same quantity, the true observed maximum; only the training label differs.

Training label	ρ , all 917	ρ , equal-effort	Enrichment, all 917	Enrichment, equal-effort
Single highest reading	0.767	0.724	4.35×	3.77×
Per-group median	0.772	0.733	4.13×	3.90×

The median leads on rank correlation in both populations and on enrichment among comparably-tested catalysts; it trails on enrichment across all 917. Differences are small and a catalyst-level bootstrap does not separate them. **We are proceeding with the per-group median**, because it cannot be set by a single high reading — a choice rather than a result, since the evidence permits but does not compel it.

6. How much of the condition grid is actually needed

Keeping only k of the ~27 measurements per catalyst–temperature cell (random draws, 5 independent repeats per k , always scored against the true maximum from the full data):

Measurements kept	Share of grid	ρ	Enrichment
1 per cell (~5 runs/catalyst)	5 %	0.759 ± 0.002	3.99×
3 per cell (~15 runs/catalyst)	15 %	0.765 ± 0.002	4.24×
all ~27 per cell	100 %	0.761	4.28×

One measurement per cell ranks almost as well as all 27, stable across draws. Labels are biased low (−2.8 yield points at $k = 1$) but roughly uniformly, so ranking survives; absolute estimates would not.

Temperature is different. 700 °C alone gives $\rho = 0.346$, enrichment 1.07×

 — no better than chance — because 70 % of catalysts peak at 800 °C or above. Redundancy is *within* a temperature, not across them: between-catalyst spread in true maximum (SD 5.32) is 3.8× the spread within one cell (SD 1.41).

This assumes the 27 slots are individually selectable — see question 2.

7. Novel promoter families, and a candidate list

Family holdouts behave reasonably for La, Ti, Zr and Ce (p 0.62–0.68) but poorly for Ba (p = 0.526, `phase4_family_diagnosis.py`). Ba catalysts average 13.76 % maximum yield against 8.95 % for the rest, and **78 % of the lab's top decile contains Ba** — removing them removes the high-yield regime. The mechanism is provable: with no Ba in training the Ba column is constant, no tree splits on it, and retraining with that column **deleted entirely gives bit-identical predictions**. The model underprices the best Ba catalysts by **9.8 yield points**.

A learning curve (`phase6_our_experiments.py`) gives a data budget: Ba runs 0.509 (none seen) → 0.565 (10) → 0.616 (25) → 0.651 (50) → 0.683 (all 204). Taking 80 % of the *final level* gives ~10 members; 80 % of the *gain* gives ~50. The second is the meaningful reading, since Ba reaches 0.509 having seen none.

We also ranked **26,414 unseen candidates** in your design grammar; `campaign_shortlist.csv` holds a suggested 17. The ranking is the deliverable, not the predicted value (ensemble maximum 18.79 % against observed yields reaching 21.50 %; catalyst-level error ~2.7 points).

8. Two questions, and what we are doing meanwhile

Both concern information absent from the exported data file that further analysis on our side cannot recover.

1. **Do the ~27 condition settings per catalyst–temperature exist in retrievable form?** They are currently invisible to the model, leaving the 19.9 % above unreachable. With them we could model conditions directly rather than averaging across them.
2. **Are those ~27 measurements distinct reaction conditions, or successive time-on-stream samples at a single condition?** We could not settle this from the file, and it changes what our target represents: under the second reading a catalyst's maximum reflects early-run behaviour rather than a sustained operating point, and §6 would not transfer to a prospective design.

A related observation rather than a question: coverage correlates with performance (Spearman(cell size, cell maximum) = +0.441) and 186 cells are absent entirely. If incomplete runs were stopped deliberately, that bias is correctable; if not, it may itself be informative. The correction in §4 holds either way.

Meanwhile, our focus is accuracy rather than scope: refining the composition-based model, since §6 suggests the data may support a simpler and more stable formulation than we use now; and revisiting the two-stage construction for cross-preparation specifically, where it still loses to a plain merge (0.318 vs 0.388). That gap looks closable.

All numbers are stored outputs of the scripts named above.